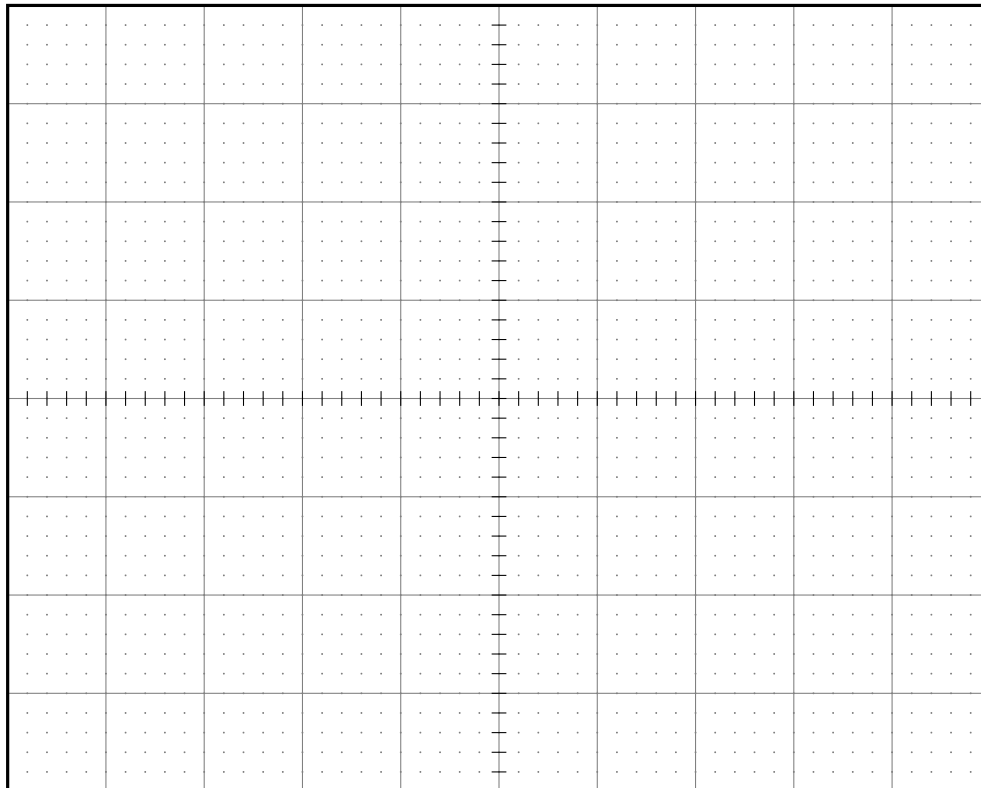


Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (20 points)*Submit the postlab to the TA at the end of the lab.***RC Circuit (9 points)****Question 3****2 points**

Known RC values: After setting up the RC circuit with $R = 3000\ \Omega$ and $C = 0.01\ \mu\text{F}$, sketch the pattern you see on the oscilloscope in Graph 1. Carefully note the value of 1 division of the time axis.

Graph 1: RC charge and discharge pattern for $R = 3000\ \Omega$ and $C = 0.01\ \mu\text{F}$.

Compute the expected value of τ below, and use the plot to estimate the time constant.

$R =$ _____ $C =$ _____ $RC =$ _____

Time of charging to within $1/e$ of max. value of V_C _____

Time of discharging to within $1/e$ of min. value of V_C _____

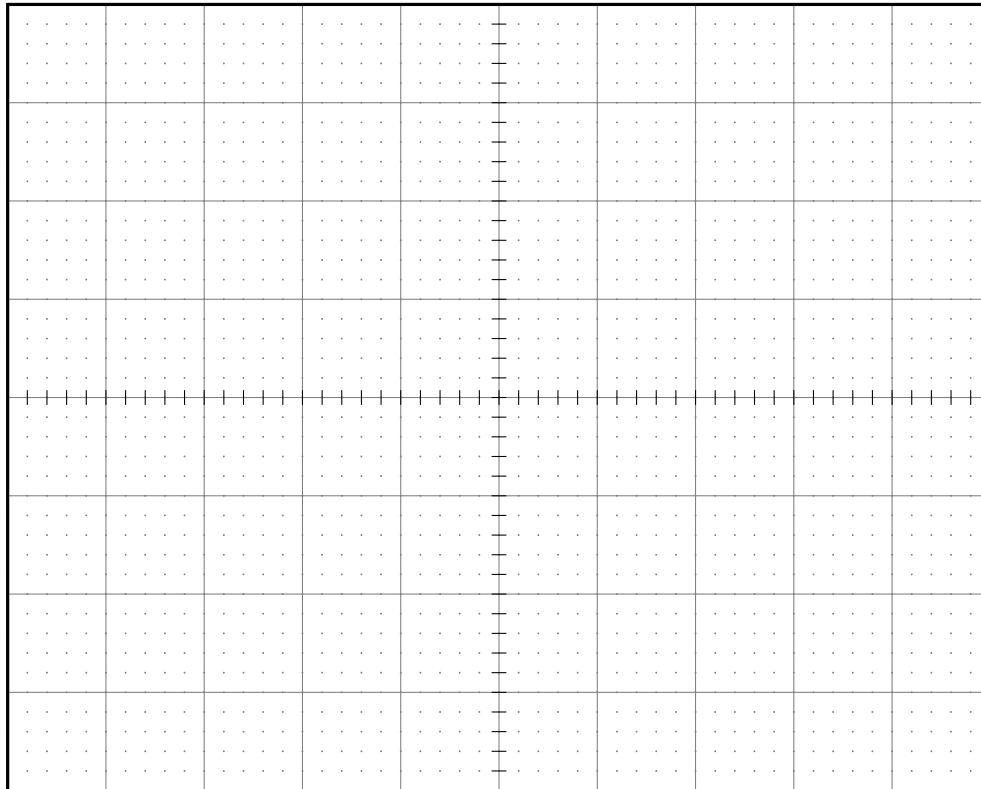
Average _____

Question 4

1 point

30% values data: change either R or C by 30% and sketch the resulting pattern in Graph 2.

Graph 2: RC charge and discharge pattern for R or C adjusted by 30%.



Compute the expected value of τ below, and use the plot to estimate the time constant.

$R =$ _____ $C =$ _____ $RC =$ _____

Time of charging to within $1/e$ of max. value of V_C _____

Time of discharging to within $1/e$ of min. value of V_C _____

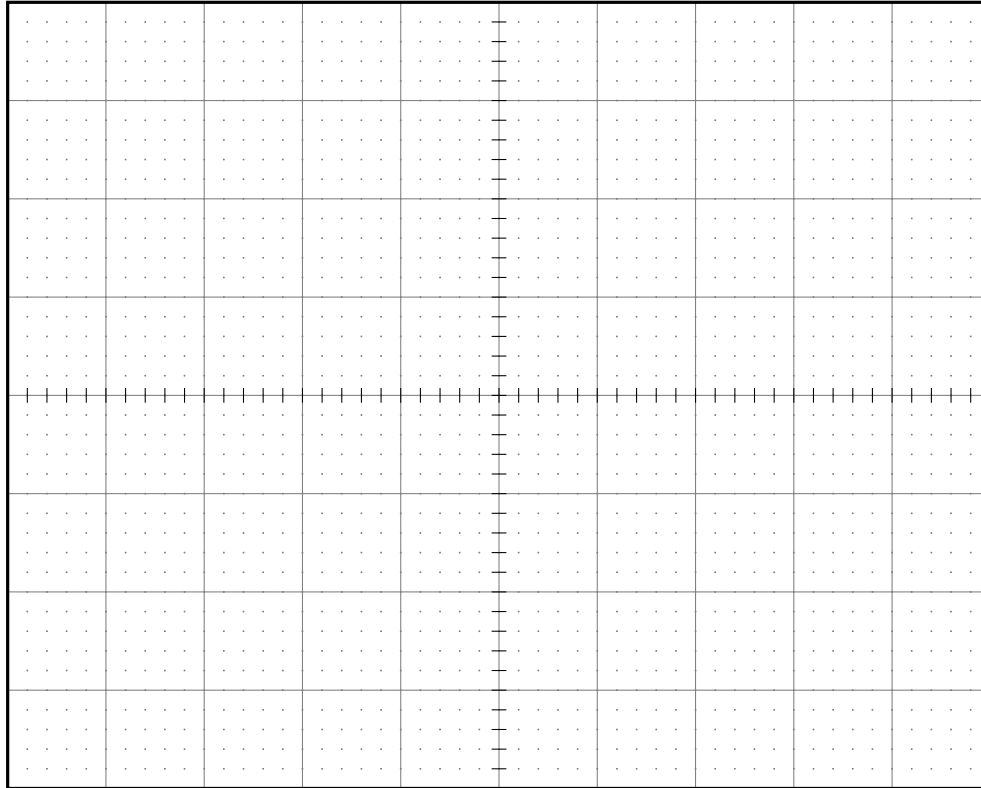
Average _____

Question 5

1 point

Unknown capacitance data:

- (a) ($1/2$ point) Swap in an unknown C and sketch the resulting pattern in Graph 3.

Graph 3: RC charge and discharge pattern for unknown capacitance C .

$R =$ _____ $C =$ **unknown** $RC =$ **unknown** _____
 Time of charging to within $1/e$ of max. value of V_C _____
 Time of discharging to within $1/e$ of min. value of V_C _____
 Average _____

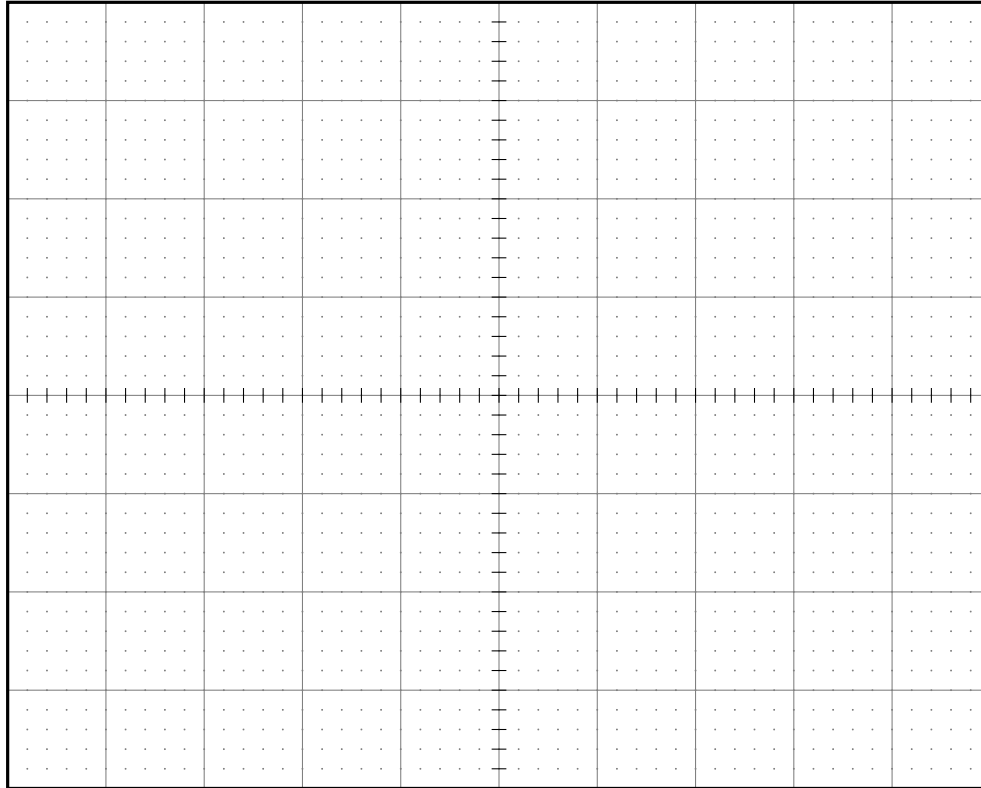
- (b) ($1/2$ point) Calculate the unknown capacitance from your measured value of RC .

Question 6

1 point

Unknown resistance data:

- (a) ($1/2$ point) Swap in an unknown R and sketch the resulting pattern in Graph 4.

Graph 4: RC charge and discharge pattern for unknown resistor R .

$R =$ unknown $C =$ _____ $RC =$ unknown
 Time of charging to within $1/e$ of max. value of V_C _____
 Time of discharging to within $1/e$ of min. value of V_C _____
 Average _____

- (b) ($1/2$ point) Calculate the unknown resistance from your measured value of RC .

Question 7**2 points**

What systematic or random errors can you think of that might account for differences between the measured and calculated values of the time constant? List at least two.

Question 8**2 points**

How accurately do you think you can measure a time constant from your traces of the oscilloscope pattern? Justify your answer.

RLC Resonant Circuit (11 points)**Question 9****3 points**

Enter your measurements in Data Table 1:

$L =$ _____ $C =$ _____ $f_0 =$ _____

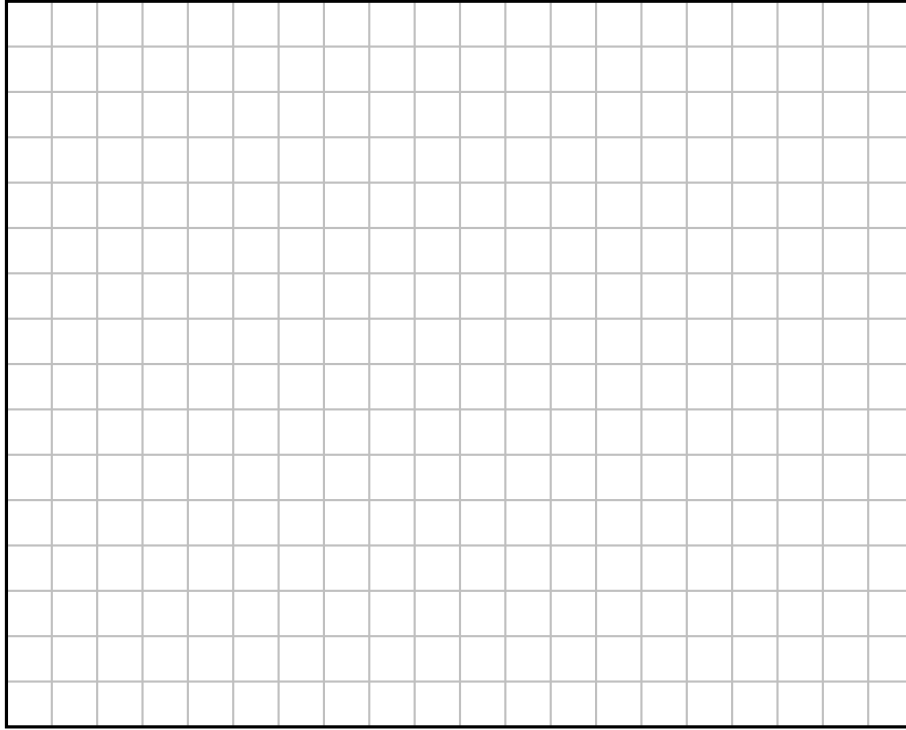
Data table 1: *Response function measurements of the RLC circuit.*

$R = 400\ \Omega$		$R = 200\ \Omega$		$R = 100\ \Omega$	
Frequency f (Hz)	Voltage V_R (V)	Frequency f (Hz)	Voltage V_R (V)	Frequency f (Hz)	Voltage V_R (V)

Question 10**3 points**

Plot resistor voltage V_R versus frequency f for all three values of R in Graph 5, using solid circles for $R = 400\ \Omega$, hollow squares for $R = 200\ \Omega$, and small x's for $R = 100\ \Omega$. Interpolate between the points in each separate set to produce smooth curves. You should find that $R = 400\ \Omega$ produces the broadest response.

Graph 5: *Response function of the RLC circuit.*

**Question 11****1 point**

How do the resonant frequencies compare in the three response functions? Is this expected? Explain.

Question 12**1 point**

From your plots, measure the Q factor for each of the three curves, showing your work for at least one of the three values. Use $Q = f_0/\Delta f$ from eq. (20) to compute Q .

Question 13**1 point**

Calculate Q for each of the three values of R used ($100\ \Omega$, $200\ \Omega$, and $400\ \Omega$) using

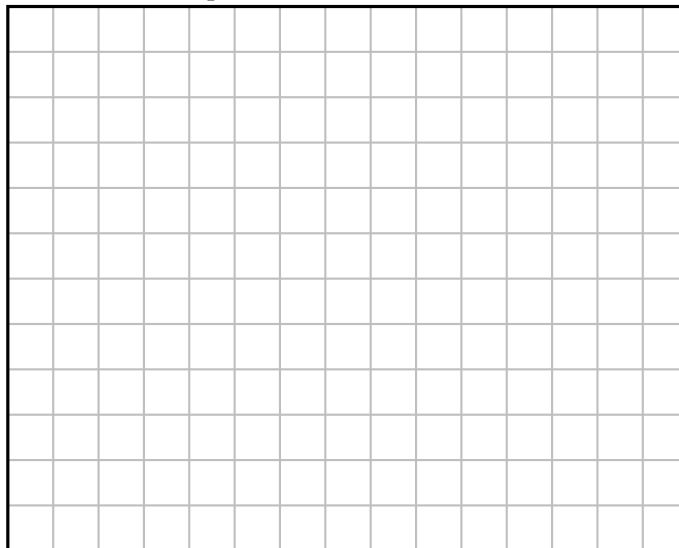
$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

from eq. (20). Show your work for at least one of the three calculations.

Question 14**2 points**

Using the values of Q that you measured from the plots of the response function, plot $1/Q$ versus R in Graph 6 below.

Graph 6: Plot of $1/Q$ versus R .



Does the plot of $1/Q$ versus R look like a straight line, as you would expect from eq. (20)? Draw a best-fit line and measure the slope. How close is your slope to its expected value?